

Unit 1-Introduction

Contents

- Data Mining and Knowledge Discovery;
- Mining Various Kinds of Knowledge;
- Data Mining Applications;
- Data Types;
- Statistics of Data;
- Similarity and Distance Measures;
- Data Quality, Data Cleaning, and Data Integration; Data Transformation;
- Dimensionality Reduction

Data Life Cycle Management

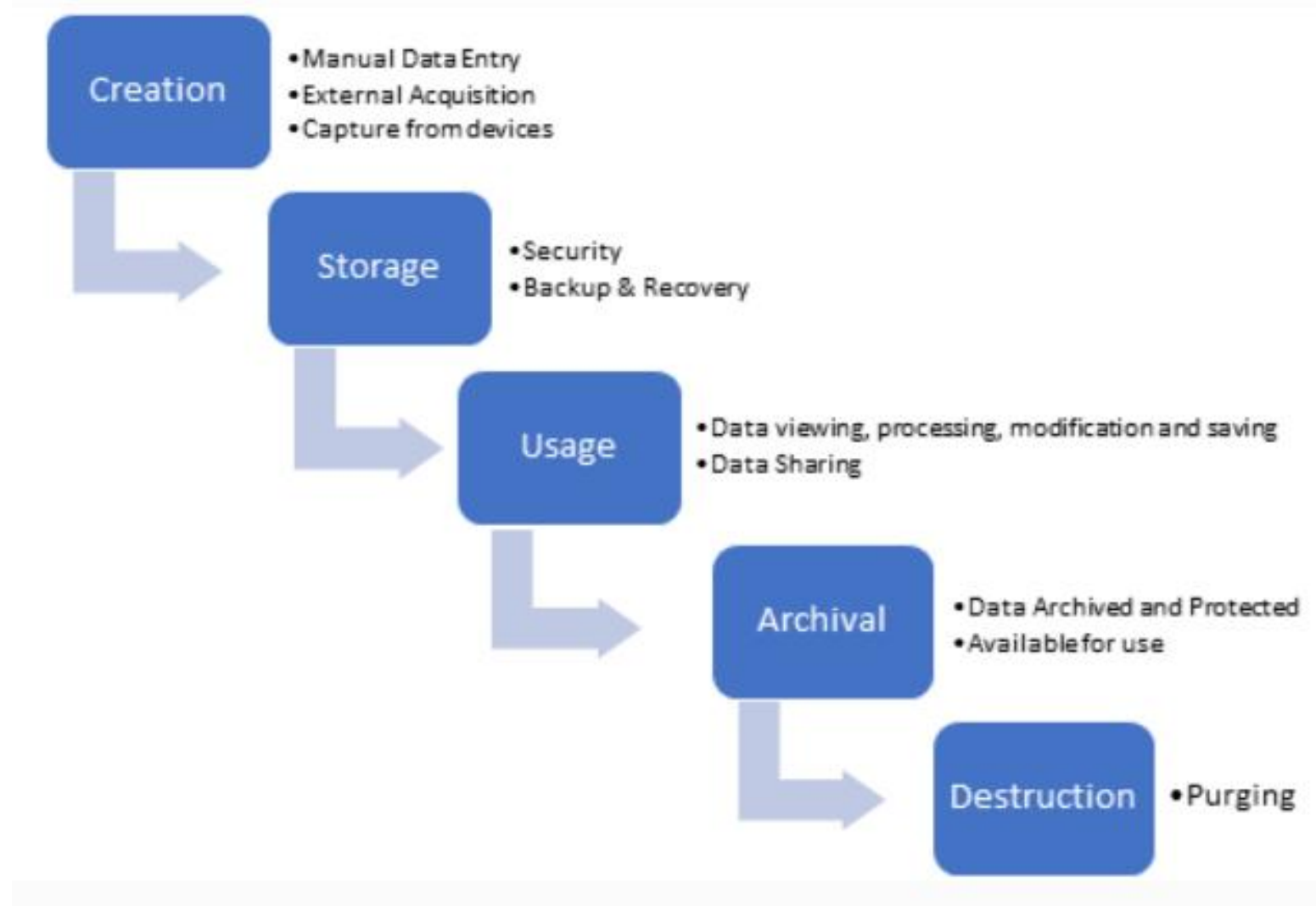


Figure: data lifecycle.

1. Data Collection/Creation

- Gather raw data from various sources for mining.
- Sources, Databases, IoT devices, social media, logs, transactions.
- For Example, A retailer collects customer purchase histories from POS systems.

2. Storage and security

- Store data securely for future mining.
- **For Examples:**
 - Storing customer purchase data in a relational database.
 - Saving user activity logs in a cloud storage service.
 - A bank encrypts transaction logs before storing them in a data warehouse.

- **Security Measures:**
 - Encryption (at rest & in transit).
 - Access controls (RBAC – Role-Based Access Control).
 - Backup & disaster recovery plans.

3. Usage

- The insights gained are used to drive decisions, processes, and strategies.
- **Audit Logs:** Track who accessed/modified data.
- **Version Control:** Maintain dataset versions for reproducibility.
- **Ethical AI:** Avoid bias in predictive models.
- For example, A healthcare provider logs all changes to patient data used in predictive diagnostics.

4. Archival

- Data that is no longer actively used but still valuable is moved to long-term storage.
- **Cold Storage:** AWS Glacier, Azure Archive (low cost, slow retrieval).
- **Data Warehouses:** Snowflake, Redshift (query able archives).
- **For Example,** A financial institution archives transaction records after 5 years but retains them for audits.

5. Destruction

- Securely delete data when no longer needed.
- **For Examples:**
 - Deleting customer data after the data retention period has expired.
 - Erasing outdated sensor logs that are no longer needed.

Types of Data

Structured Data

- Structured data is the easiest to search, organize, and analyze because it is arranged in a predefined manner-usually in rows and columns.
- Each element can be mapped into fixed, pre-defined fields.
- **For Example:**
 - Think about data stored in an Excel spreadsheet — like sales records categorized by region, product, or customer.
 - Structured data usually follows a data model designed by a database architect and is managed using **Structured Query Language (SQL)**.

Semi-Structured Data

- Semi-structured data falls between structured and unstructured data. It contains both organized fields and free-form elements.
- **For Examples:**
- Email messages are a good example. The email body (content) is unstructured, but there are structured parts like the sender's name, recipient's address, timestamp, and subject line.
- Other examples include **XML files**, **JSON files**, and **NoSQL databases**.

Unstructured Data

- A much larger percentage of all the world's data is unstructured.
- Unstructured data cannot be easily contained within a traditional row-column database and has no associated formal data model.

- **For Examples:**
 - **Text:** Email bodies, PDFs, Word documents, open-ended survey responses.
 - **Media:** Photos, videos, audio recordings.
 - **Web Content:** Social media posts, websites, blogs.
 - **Other Sources:** Satellite imagery, call center transcripts/recordings, presentations.

How Data Mining Handles Unstructured Data?

- **Text Mining:** NLP techniques (tokenization, topic modeling) for emails/social media.
- **Computer Vision:** Object detection in images/videos (e.g., facial recognition).
- **Audio Mining:** Speech-to-text conversion for call center analytics.
- **Metadata Extraction:** Leveraging tags (geotags, timestamps) for patterns.

Structured data



Characteristics

Predefined data models
Easy to search
Text-based
Shows what's happening

Resides in

Relational databases
Data warehouses

Stored in

Rows and columns

Examples

Dates, phone numbers, social security numbers, customer names, transaction info

Unstructured data



Characteristics

No predefined data models
Difficult to search
Text, pdf, images, video
Shows the why

Resides in

Applications
Data warehouses and lakes

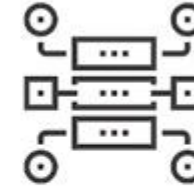
Stored in

Various forms

Examples

Documents, emails and messages, conversation transcripts, image files, open-ended survey answers

Semi-structured data



Characteristics

Loosely organized
Meta-level structure that can contain unstructured data
HTML, XML, JSON

Resides in

Relational databases
Tagged-text format

Stored in

Abstracts & figures

Examples

Server logs, tweets organized by hashtags, emails sorting by folders (inbox; sent; draft)

Data Mining

- **Data mining is the process of discovering interesting patterns, models, and other kinds of knowledge in large data sets**, using artificial intelligence, machine learning, statistics, and database systems
- The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.
- Data mining is considered as a synonym for another popularly used term, known as **KDD, knowledge discovery in databases**.
- **Data mining** is the process of discovering interesting patterns and knowledge from massive amounts of data. As a *knowledge discovery process*, it typically involves data cleaning, data integration, data selection, data transformation, pattern and model discovery, pattern or model evaluation, and knowledge presentation.

The main goal of the data mining process is to extract information from a data set and transform it into an understandable structure for further use.

Data Mining: an essential step in knowledge discovery

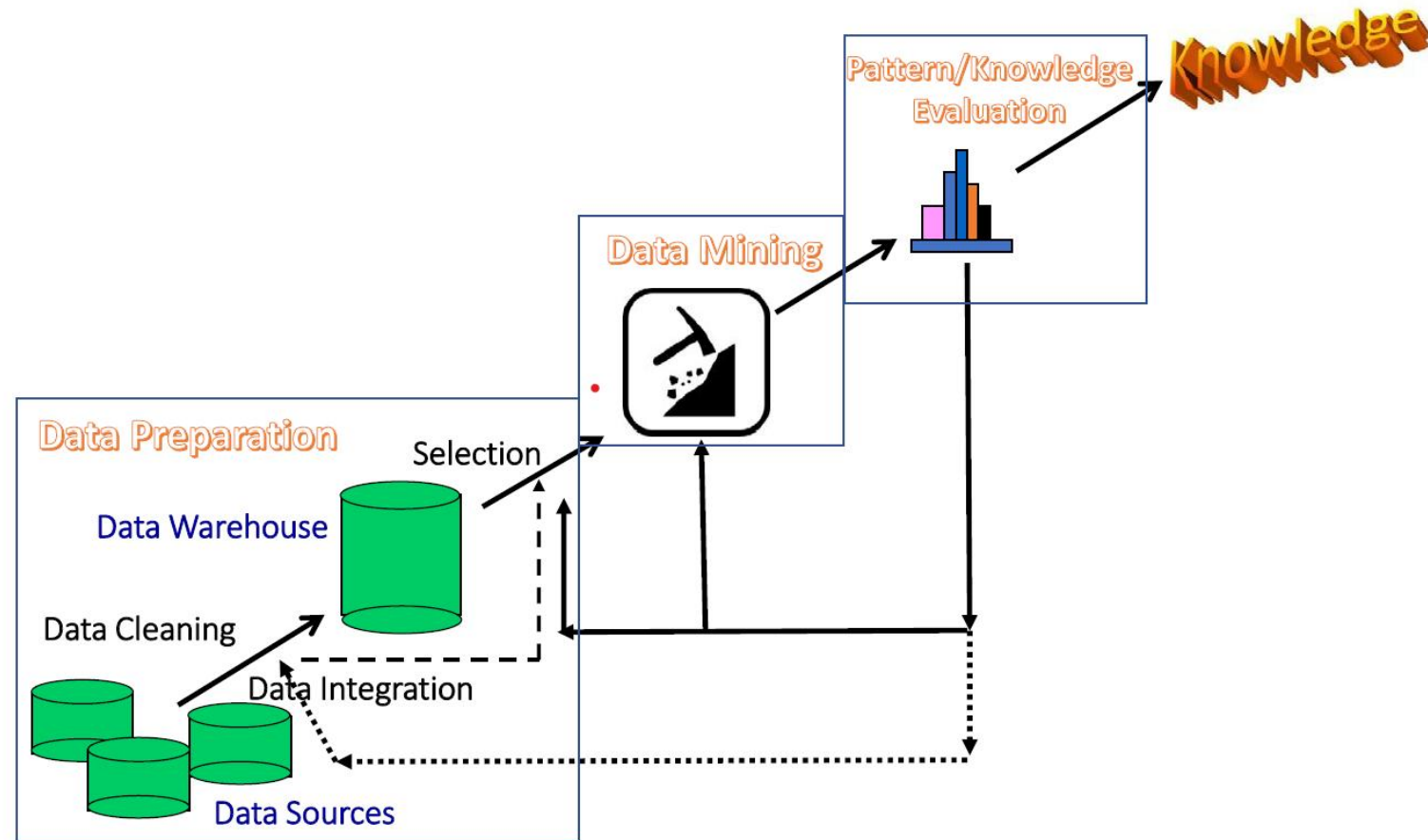


Figure: An essential step in the process of knowledge discovery

- Many people treat data mining as a synonym for another popularly used term, **knowledge discovery from data**, or **KDD**, whereas others view data mining as merely an essential step in the overall process of knowledge discovery.

1. Data preparation

- a. **Data cleaning** (to remove noise and inconsistent data)
- b. **Data integration** (where multiple data sources may be combined)
- c. **Data transformation** (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
- d. **Data selection** (where data relevant to the analysis task are retrieved from the database)

2. Data mining (an essential process where intelligent methods are applied to extract patterns or construct models)

3. Pattern/model evaluation (to identify the truly interesting patterns or models representing knowledge based on *interestingness measures*)

4. Knowledge presentation (where visualization and knowledge representation techniques are used to present mined knowledge to users)

Data Mining Functionalities

- Data mining functionalities are used to specify the kind of patterns to be found in data mining tasks.
- Data mining tasks can be classified into two categories: **descriptive and predictive**.
- **Descriptive mining** tasks characterize the general properties of the data in the database. The common data features are highlighted in the data set. E.g., count, average etc.
- **Predictive mining** tasks perform inference on the current data in order to make predictions.
- For example, predicting the volume of business next quarter based on performance in the previous quarters over several years
- There are various data mining functionalities which are as follows –

1. Class/Concept Descriptions

- Data can be associated with classes or concepts.
- For example, in the Electronics store, classes of items for sale include computers and printers, and concepts of customers include big Spenders and budget Spenders.
- There are two concepts here, one that helps with grouping and the other that helps in differentiating.
- **Data Characterization:** This refers to the summary of general characteristics or features of the class, resulting in specific rules that define a target class. A data analysis technique called Attribute-oriented Induction is employed on the data set for achieving characterization.
- **Data Discrimination:** Discrimination is used to separate distinct data sets based on the disparity in attribute values. It compares features of a class with features of one or more contrasting classes e.g., bar charts, curves and pie charts.

2. Mining Frequent Patterns

One of the functions of data mining is finding data patterns. Frequent patterns are things that are discovered to be most common in data. Various types of frequency can be found in the dataset.

- **Frequent item set:** This term refers to a group of items that are commonly found together, such as milk and sugar.
- **Frequent substructure:** It refers to the various types of data structures that can be combined with an item set or subsequences, such as trees and graphs.
- **Frequent Subsequence:** A regular pattern series, such as buying a phone followed by a cover.

3. Association Analysis

- It analyses the set of items that generally occur together in a transactional dataset. It is also known as Market Basket Analysis for its wide use in retail sales. Two parameters are used for determining the association rules:
 - It provides which identifies the common item set in the database.
 - Confidence is the conditional probability that an item occurs when another item occurs in a transaction.

- Suppose, as a marketing manager, you would like to determine which items are frequently purchased together within the same transactions.

buys(X, “computer”) = buys(X, “software”) [support=1%, confidence=50%]

Where X is a variable representing a customer. Confidence=50% means that if a customer buys a computer, there is a 50% chance that she will buy software as well. Support=1% means that 1% of all of the transactions under analysis showed that computer and software were purchased together.

4. Classification

- Classification is a data mining technique that categorizes items in a collection based on some predefined properties.
- It uses methods like if-then, decision trees or neural networks to predict a class or essentially classify a collection of items.
- A training set containing items whose properties are known is used to train the system to predict the category of items from an unknown collection of items.

5. Prediction

- It defines predict some unavailable data values or spending trends.
- It can be a prediction of missing numerical values or increase or decrease trends in time-related information. There are primarily two types of predictions in data mining: numeric and class predictions.
 - **Numeric predictions** are made by creating a linear regression model that is based on historical data.
 - Prediction of numeric values helps businesses ramp up for a future event that might impact the business positively or negatively.
 - **Class predictions** are used to fill in missing class information for products using a training data set where the class for products is known.

6. Cluster Analysis

- It is similar to classification, but the classes are not predefined. Data attributes represent the classes. Similar data are grouped together, with the difference being that a class label is not known.
- Clustering algorithms group data based on similar features and dissimilarities.
- In image processing, pattern recognition and bioinformatics, clustering is a popular data mining functionality.

7. Outlier Analysis

- Outlier analysis is important to understand the quality of data. If there are too many outliers, you cannot trust the data or draw patterns.
- Outliers are data elements that cannot be grouped in a given class or cluster. These are the data objects which have multiple behavior from the general behavior of other data objects.
- An outlier analysis of the data that cannot be grouped into any classes by the algorithms is pulled up.

8. Evolution and Deviation Analysis

- Evolution Analysis pertains to the study of data sets that change over time.
- Evolution analysis models are designed to capture evolutionary trends in data helping to characterize, classify, cluster or discriminate time-related data.

9. Correlation Analysis

- Correlation is a mathematical technique for determining whether and how strongly two attributes is related to one another.
- It refers to the various types of data structures, such as trees and graphs, that can be combined with an item set or subsequence.

Data Mining Applications

- **Business intelligence**

- It is critical for businesses to acquire a better understanding of the commercial context of their organization, such as their customers, the market, supply and resources, and competitors.
- **Business intelligence (BI)** technologies provide historical, current, and predictive views of business operations.
- **For examples,** include reporting, online analytical processing, business performance management, competitive intelligence, benchmarking, and predictive analytics.
- **For Examples, Market Basket Analysis:** Discovering which products are frequently bought together (e.g., customers who buy bread often buy butter).
- **Customer Segmentation:** Grouping customers based on purchasing behavior to tailor marketing strategies.
- **Sales Forecasting:** Predicting future sales based on historical data trends.

“How important is data mining in business intelligence?”

- Without data mining, many businesses may not be able to perform effective market analysis, compare customer feedback on similar products, discover the strengths and weaknesses of their competitors, retain highly valuable customers, and make smart business decisions.
- Techniques like **clustering**, **classification**, and **outlier analysis** are commonly used for customer segmentation, risk management, and targeted marketing.

- **Web Search Engines**

- **Search engines** heavily rely on data mining to improve the quality of search results and advertisements.
- Data mining is used to analyze **clickstreams**, **query logs**, and **user behavior** to rank pages and suggest relevant content.
- For Example, Google uses techniques like **link analysis** (e.g., PageRank), **clickstream mining**, and **text mining** to rank websites based on user relevance and behavior.
- Tasks like **classification**, **clustering**, and **recommendation** models are employed for personalized search experiences.

- **Social Media and Social Networks**

- **Social media mining** involves analyzing massive amounts of social media data to identify patterns and trends related to online behavior, connections, and content exchange.
- It supports applications like social event detection, public health monitoring, sentiment analysis, recommendation systems, trustability analysis, and spammer detection.
- **Social network mining** focuses on studying the structure and information of social networks using graph theory and data mining techniques.
- It analyzes nodes (individuals or entities) and ties (relationships) to uncover hidden communities, network dynamics, information propagation, influence modeling and conduct location-based social network analysis.

- Social networks such as **Facebook, Twitter, and LinkedIn** generate vast amounts of **semi-structured and unstructured data**.
- Data mining helps in understanding **user interests, community detection, sentiment analysis, and influencer identification**.
- Both social media and social network mining are vital applications of data mining for understanding online interactions and behaviors.

- **Biology, medical science, and health care**

- Biology, medical science, and healthcare are producing massive amounts of data from sources like **genomics, imaging, mobile health, and electronic health records**.
- The rapid growth of biomedical data demands new strategies for management and analysis to advance scientific understanding and healthcare improvements.
- Biomedical data mining addresses challenges such as mining genomic and proteomic sequences, detecting subgraph patterns, analyzing regulatory networks, predicting protein-protein interactions, classifying medical images, mining biological texts, constructing biological information networks, and analyzing electronic health records and biomedical networks.

Data objects and Attribute Types

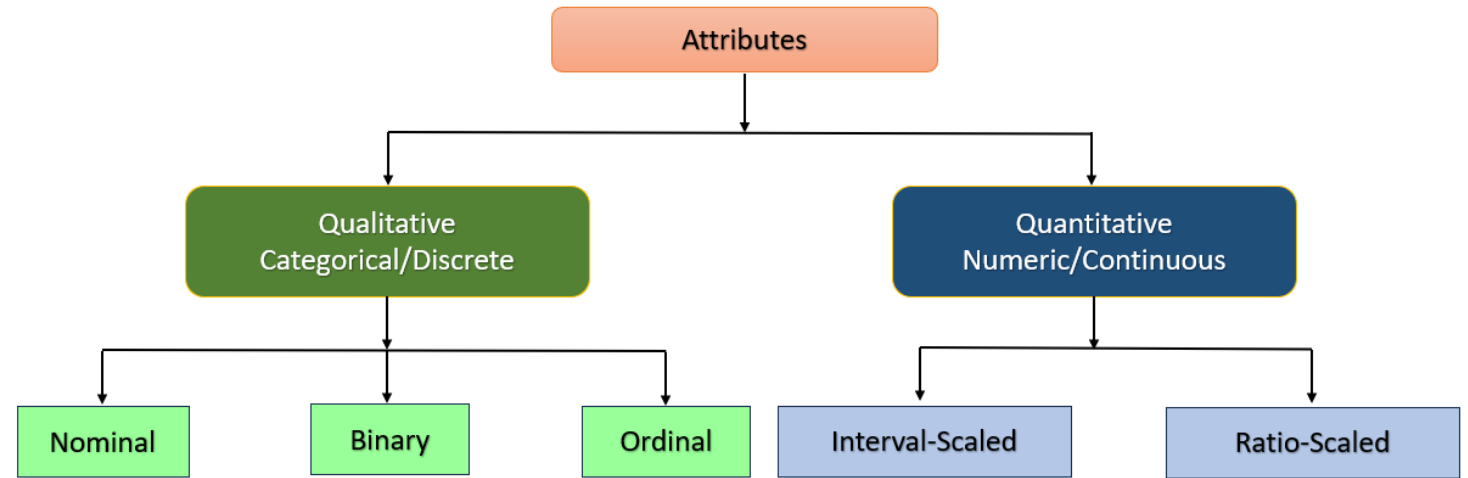
- Data sets are made up of data objects.
- A **data object** represents an entity—in a **sales database**, the objects may be customers, store items, and sales;
- **in a medical database**, the objects may be patients;
- **In a university database**, the objects may be students, professors, and courses.
- Data objects are typically described by attributes. Data objects can also be referred to as **samples, examples, instances, data points, or objects**.
- If the data objects are stored in a database, they are *data tuples*. That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.

What is an attribute?

- An **attribute** is a data field, representing a characteristic or feature of a data object.
- The nouns **attribute**, **dimension**, **feature**, and **variable** are often used interchangeably in the literature.
- The term **dimension** is commonly used in data warehousing.
- Machine learning literature tends to use the term **feature**, whereas statisticians prefer the term **variable**.
- Data mining and database professionals commonly use the term **attribute**.
- Attributes describing a customer object can include, for example, customer_ID, name, and address.
- Observed values for a given attribute are known as **observations**. A set of attributes used to describe a given object is called an **attribute vector(or feature vector)**.

Types of Attributes:

- Nominal Attributes
- Binary Attributes
- Ordinal Attributes
- Numeric Attributes
- Discrete versus Continuous Attributes



Nominal attributes

- Nominal means “relating to names.” The values of a **nominal attribute** are symbols or **names of things**.
- Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as **categorical**.
- The values do not have any meaningful order. In computer science, the values are also known as **enumerations**.
- **For Examples;**

<u>Attributes</u>	<u>Possible Values</u>
Hair_color :	black, brown, red, green, and so on.
Marital_status :	single, married, divorced, and widowed.
Occupation :	teacher, doctor, farmer, student and so on.
Blood type :	O negative, O positive, A negative, and so on

Binary attributes

- A **binary attribute** is a nominal attribute with only two categories or states: 0 or 1, where 0 typically means that the attribute is absent, and 1 means that it is present.
- A binary attribute is **symmetric** if both of its states are equally valuable and carry the same weight; i.e., there is no preference on which outcome should be coded as 0 or 1.
- For example: **Gender** with states **male** and **female**.
- A binary attribute is **asymmetric** if the outcomes of the states are not equally important.
- For example: **Medical test for TB** (Tuberculosis): **1** - more important/rarer case (e.g., **TB+**). **0** - more common case (e.g., **TB -**).

Ordinal Attribute

- An **ordinal attribute** is an attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known.
- For example, `drink_size` corresponds to the size of drinks available at a fast-food restaurant. This nominal attribute has three possible values: **small**, **medium**, and **large**.
- The values have a meaningful sequence (which corresponds to increasing drink size);
- **For customer Feedback:**
 - Values: *Very Unsatisfied, Unsatisfied, Neutral, Satisfied, Very Satisfied*
 - Order: *Very Satisfied > Satisfied > Neutral > Unsatisfied > Very Unsatisfied*

- **For Job Titles:**
 - Values: Intern ,Junior Developer, Senior Developer, Team Lead ,Manager
 - Job rank: Intern < Junior Developer < Senior Developer < Team Lead < Manager

Numeric attributes

- A **numeric attribute** is quantitative; that is, it is a measurable quantity, represented in integer or real values. Numeric attributes can be interval-scaled or ratio-scaled.

Interval-scaled attributes:

- **Interval-scaled attributes** are measured on a scale of equal-size units.
- The values of interval-scaled attributes have order and can be positive, 0, or negative.
- Thus, in addition to providing a ranking of values, such attributes allow us to compare and quantify the difference between values.
- For example, A **temperature** attribute is interval-scaled, for a number of different days, each day is an object. By ordering the values, we obtain a ranking of the objects with respect to **temperature**, 20°C is 5°C higher than 15°C.
- **Calendar dates** are another example. For instance, the years 2012 and 2020 are eight years apart.

- Temperatures in Celsius do not have a true zero point, that is, 0°C does not indicate “no temperature.”
- Calendar dates do not have a true zero point, that is, the year 0 not the beginning of the time.

Ratio-scaled attributes:

- A **ratio-scaled attribute** is a numeric attribute with an inherent zero-point. i.e., if a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value.
- In addition, the values are ordered, and we can also compute the difference between values, as well as the mean, median, and mode.
- For examples, ratio-scaled attributes include **count** attributes such as *years_of_experience* (e.g., the objects are employees) and *number_of_words* (e.g., the objects are documents).
- Physical measurements (e.g., weight, height, speed) and monetary quantities (e.g., \$100 is truly 100 times more than \$1).

Discrete Attributes

- A **discrete attribute** has a **finite** or **countably infinite** set of values, which may or may not be represented as integers.
- They can be **numerical** (e.g., number of children in a family: 0, 1, 2, ...).) or **categorical** (e.g., blood type: A, B, AB, O).
- For examples: -Attributes(values)
 - Students (25, 30, 35 ...etc)
 - ZIP Code (44600, 11040, ... etc)
 - Colors (red, blue, green.. etc)

Continuous Attributes:

- Continuous data have infinite no of states. Continuous attributes are typically represented as floating-point variables.
- Fo examples: Height (165.3 cm, 170.8 cm.. etc)
- Temperature (23.7°C, 36.5°C... etc)

Statistics of data

- For data preprocessing tasks, we want to learn about data characteristics regarding **central tendency** of the data.
- Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers.
- **This section discusses three areas of basic statistical descriptions, those are:**
 - **Measuring the Central Tendency:** Mean, Median, and Mode
 - **Measuring the Dispersion of Data:** Range, Quartiles, Variance, Standard Deviation, and Interquartile Range (IQR)
 - Graphic Displays of Basic Statistical Descriptions of to visually inspect our data.

Measuring the Central Tendency: Mean, Median, and Mode:

- The central tendency is stated as the statistical measure that represents the single value of the entire distribution or a dataset. It aims to provide an accurate description of the entire data in the distribution.
- Suppose that we have some attribute X , like salary, which has been recorded for a set of objects. Let $x_1, x_2, \dots, x_N$ be the set of N observed values or *observations* for X . Here, these values may also be referred to as the data set (for X).
- If we were to plot the observations for *salary*, where would most of the values fall? This gives us an idea of the central tendency of the data. Measures of central tendency include the mean, median, mode, and midrange

Mean

- Mean is the average of all data values which you work with.

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}.$$

- Sometimes, each value x_i in a set may be associated with a weight w_i for $i = 1, \dots, N$. The weights reflect the significance, importance, or occurrence frequency attached to their respective values.

Weighted arithmetic mean or the **weighted average**

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_N x_N}{w_1 + w_2 + \cdots + w_N}.$$

Eg, GPA Calculation

- Math (4 credits) = 90%
- English (3 credits) = 80%
- Science (2 credits) = 85%

Weighted GPA

$$((90*4)+(80*3)+(85*2))/(4+3+2)$$

Median:

- Suppose that a given data set of N distinct values is **sorted** in numerical order.
 - If N is **odd**, the median is the **middle value** of the ordered set;
 - If N is **even**, the median is the **average of the middle two values**.
- Suppose the following values for salary (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

There is an **even number** of observations (i.e., 12); therefore, the median is not unique. It can be any value within the two middlemost values of 52 and 56 (i.e., within the 6th and 7th values in the list). By convention, we assign the **average of the two middlemost values** as the median;

=> $(52+56)/2 = 54$ Thus, the median is \$54,000.

- In probability and statistics, the median generally applies to numeric data; however, we may extend the concept to ordinal data.
- Suppose that a given data set of N values for an attribute X is sorted in increasing order.
 - If **N is odd**, then the median is the **middle value** of the ordered set.
 - If **N is even**, then the **median may not be not unique**.
- In this case, the median is the two middlemost values and any value in between.

Mode:

- The mode for a set of data is the value that occurs **most frequently** in the set.
- It is possible for the greatest frequency to correspond to several different values, which results in more than one mode.
 - Data sets with one, two, or three modes: called **unimodal**, **bimodal**, and **trimodal**.
 - At the other extreme, if each data value occurs **only once**, then there is **no mode**.
- Suppose we have the following values for salary (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

The above data has **bimodal mode**. i. e. The two modes are 52 and 70.

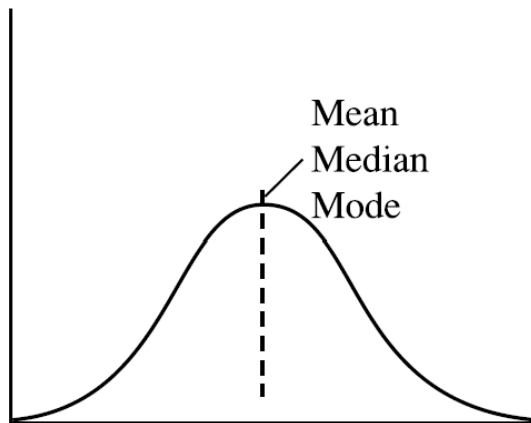
Midrange:

- It is the **average of the largest and smallest values** in the set.
- Suppose we have the following values for salary (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110. The midrange of the data is $30,000 + 110,000 / 2 = \$70,000$ Thus, the median is \$70,000

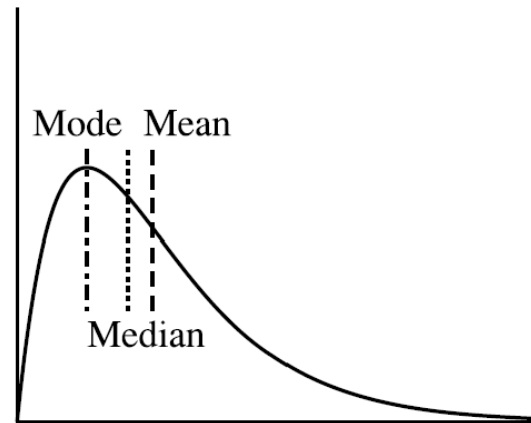
Central Tendency Measures for different attributes:

- Central Tendency Measures for Numerical Attributes: **Mean, Median, Mode**
- Central Tendency Measures for Categorical Attributes:
 - Central Tendency Measures for Nominal Attributes: **Mode**
 - Central Tendency Measures for Ordinal Attributes: **Mode, Median**

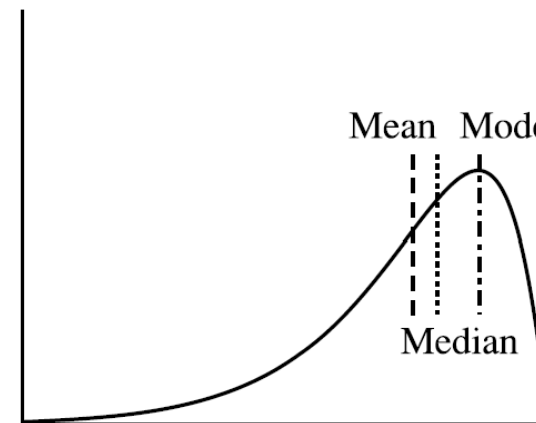
- In a unimodal frequency curve with perfect symmetric data distribution, the mean, median, and mode are all at the same center value, as shown in (fig. a).
- Data in most real applications are not symmetric. They may instead be either positively skewed, where the mode occurs at a value that is smaller than the median (Fig. b), or
- Negatively skewed, where the mode occurs at a value greater than the median (fig. c).



(a) Symmetric data



(b) Positively skewed data



(c) Negatively skewed data

Measuring the dispersion of data: Range, Quartiles, Variance, Standard Deviation, and Interquartile

- The main purpose of measures of dispersion is to get as much as possible a true picture of the data. Dispersion indicates the spread or variability of the data in a distribution.
- In other words, dispersion is the amount of spread of data about the center of the distribution.

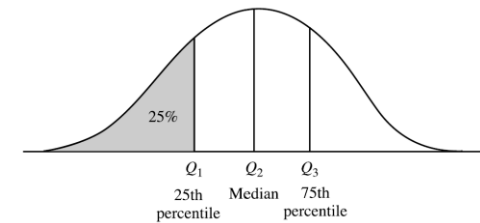
Range:

- To calculate a range, prepare all the values in ascending order, then subtract the lowest value from the highest value.

$$\text{Range} = \text{Highest_value} - \text{Lowest_value}$$

Quantiles :

- Suppose that the data for attribute X are sorted in ascending numeric order. Imagine that we can pick certain data points so as to split the data distribution into equal-size consecutive sets.
- These data points are called *quantiles*. **Quantiles** are points taken at regular intervals of a data distribution, dividing it into essentially equal-size consecutive sets.
- The 4-quantiles are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution called **quartiles**.
- The three quartiles divide the distribution into four equal-size consecutive subsets. The second quartile corresponds to the median.



- The 100-quantiles are more commonly referred to as **percentiles**; they divide the data distribution into 100 equal-size consecutive sets.
- The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data. This distance is called the **interquartile range (IQR)** and is defined as

$$\text{IQR} = Q3 - Q1.$$

- Suppose we have the following values for salary (in thousands of dollars), shown in increasing order:
30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110. (12 observations)

Thus, the quartiles for this data are the third, sixth, and ninth values, respectively, in the sorted list.

Therefore, $Q1 = \$47,000$ and $Q3$ is $\$63,000$.

Thus, the interquartile range is $\text{IQR} = 63 - 47 = \$16,000$.

(Note that the sixth value is a median, $\$52,000$, although this data set has two medians since the number of data values is even.).

Variance and standard deviation:

- Variance measure the variability of the data around its mean or average.
- High variance means there is more variability or we can say that the data deviates more from its mean whereas low variance means there is less variability.
- If the variance is zero that means all the values in the data are identical. Variance can never be negative.

$$\text{(Variance) } \sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

- Variance and standard deviation are measures of data dispersion. They indicate how spread out a data distribution is. A low standard deviation means that the data observations tend to be very close to the mean, whereas a high standard deviation indicates that the data are spread out over a large range of values.
- The **variance** of N observations, $x_1, x_2, \dots, x_N$ (when N is large), for a numeric attribute X is,

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2,$$

- Standard deviation is simply the square root of variance. It indicates how far away the dispersion of the dataset from its mean. It is denoted by (sigma).
- Simply standard deviation helps us to find the spread of the data about its mean or average.
- A low Standard deviation indicates that the data are less spread from their average where a high standard deviation indicates the data are more spread out from its average.

$$\text{(Standard Deviation) } \sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

Graphic Displays of Basic Statistical Descriptions of Data

- Quantile Plots,
 - Quantile–Quantile Plots,
 - Histograms, and Scatter Plots.
- Such graphs are helpful for the visual inspection of data, which is useful for data preprocessing.

Quantile Plots:

- A **quantile plot** is a simple and effective way to have a first look at a univariate data distribution. First, it displays all of the data for the given attribute (allowing a user to assess both the overall behavior and unusual occurrences).
- Note that the 0.25 quantile corresponds to quartile $Q1$, the 0.50 quantile is the median, and the 0.75 quantile is $Q3$.

$$f_i = \frac{i - 0.5}{N}.$$

Table 2.3 A set of unit price data for items sold at a branch of the online store.

Unit price (\$)	Count of items sold
40	275
43	300
47	250
⋮	⋮
74	360
75	515
78	540
⋮	⋮
115	320
117	270
120	350

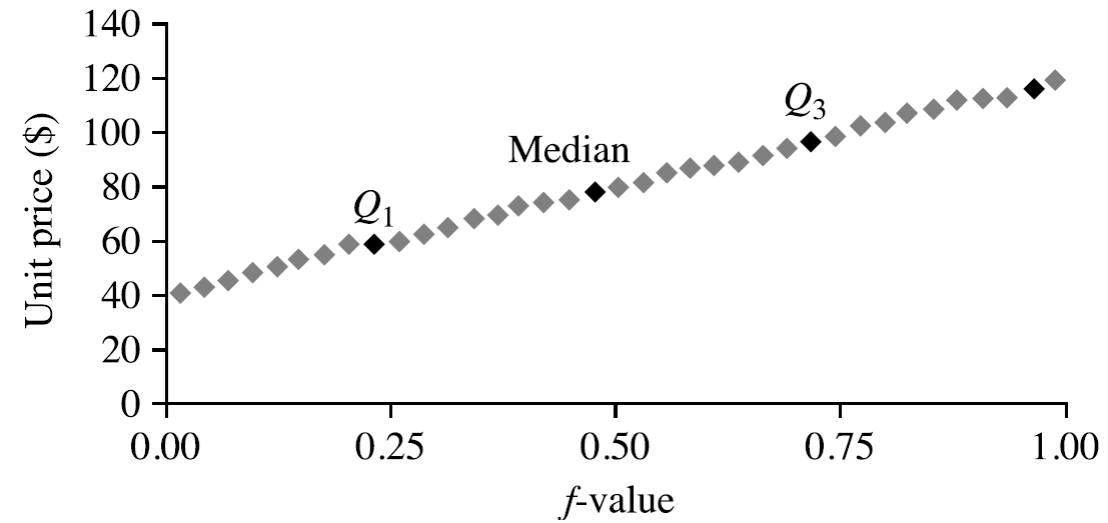


Fig: A quantile plot for the unit price data of table.

Quantile-quantile plot:

- A **quantile-quantile plot**, or **q-q plot**, graphs the quantiles of one univariate distribution against the corresponding quantiles of another.
- It is a powerful visualization tool in that it allows the user to view whether there is a shift in going from one distribution to another.
- Suppose that we have two sets of observations for the attribute or variable *unit price*, taken from two different branch locations.
- Let $x_1, \dots, x_N$ be the data from the first branch, and $y_1, \dots, y_M$ be the data from the second, where each data set is sorted in ascending order.
- If $M = N$ (i.e., the number of points in each set is the same), then we simply plot y_i against x_i , where y_i and x_i are both $(i - 0.5)/N$ quantiles of their respective data sets.

- If $M < N$ (i.e., the second branch has fewer observations than the first), there can be only M points on the q-q plot. Here, y_i is the $(i - 0.5)/M$ quantile of the y data, which is plotted against the $(i - 0.5)/M$ quantile of the x data. This computation typically involves interpolation.
- **For Example,** Fig. shows a quantile-quantile plot for *unit price* data of items sold at two branches of the online store during a given time period. Each point corresponds to the same quantile for each data set and shows the unit price of items sold at branch 1 vs. branch 2 for that quantile.
- We see, for example, that at $Q1$, the unit price of items sold at branch 1 was slightly less than that at branch 2. In other words, 25% of items sold at branch 1 were less than or equal to \$60, whereas 25% of items sold at branch 2 were less than or equal to \$64. At the 50th percentile (marked by the median, which is also $Q2$),

- We see that 50% of items sold at branch 1 were less than \$78, whereas 50% of items at branch 2 were less than \$85.
- In general, we note that there is a shift in the distribution of branch 1 with respect to branch 2 in that the unit prices of items sold at branch 1 tend to be lower than those at branch 2.

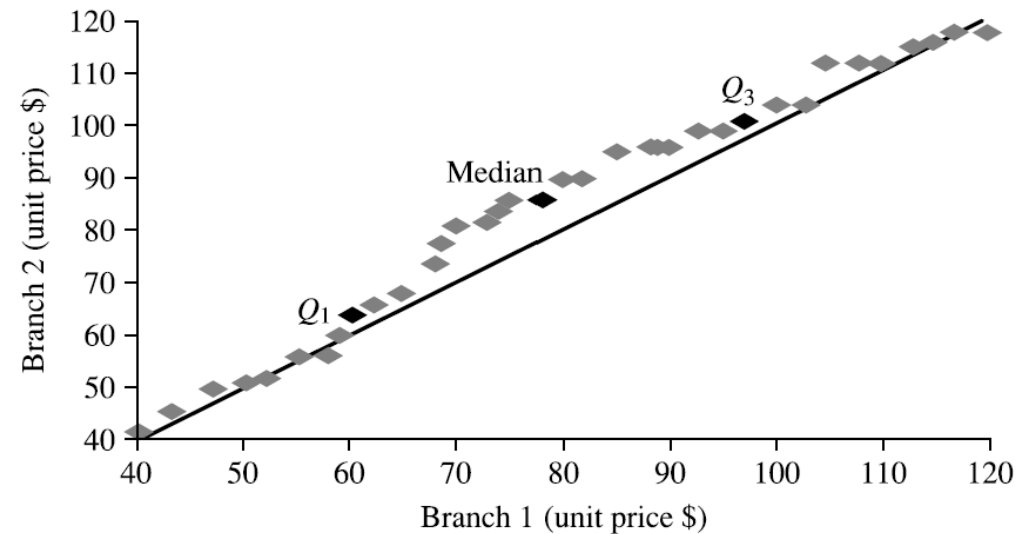


Fig: A q-q plot for unit price data from two branches of the online store.

Histograms:

- **Histograms** (or **frequency histograms**) are at least a century old and are widely used. “Histos” means pole or mast, and “gram” means chart, so a histogram is a chart of poles.
- Plotting histograms is a graphical method for summarizing the distribution of a given attribute, X . According to the number of poles desired in the chart, the range of values for X is partitioned into a set of disjoint consecutive subranges.
- For example, a price attribute with a value range of \$1–\$200 (rounded up to the nearest dollar) can be partitioned into subranges 1–20, 21–40, 41–60, and so on. For each subrange, a bar is drawn with a height that represents the total count of items observed within the subrange.

Scatter plots and data correlation:

- A **scatter plot** is one of the most effective graphical methods for determining whether there appears to be a relationship, pattern, or trend between two numeric attributes.
- To construct a scatter plot, each pair of values is treated as a pair of coordinates in an algebraic sense and plotted as points in the plane.
- The scatter plot is a useful method for providing a first look at bivariate data to see clusters of points and outliers, or to explore the possibility of correlation relationships. Two attributes, X and Y , are **correlated** if the knowledge of one attribute enables to predict the other with some accuracy.

Table 2.3 A set of unit price data for items sold at a branch of the online store.	
Unit price (\$)	Count of items sold
40	275
43	300
47	250
⋮	⋮
74	360
75	515
78	540
⋮	⋮
115	320
117	270
120	350

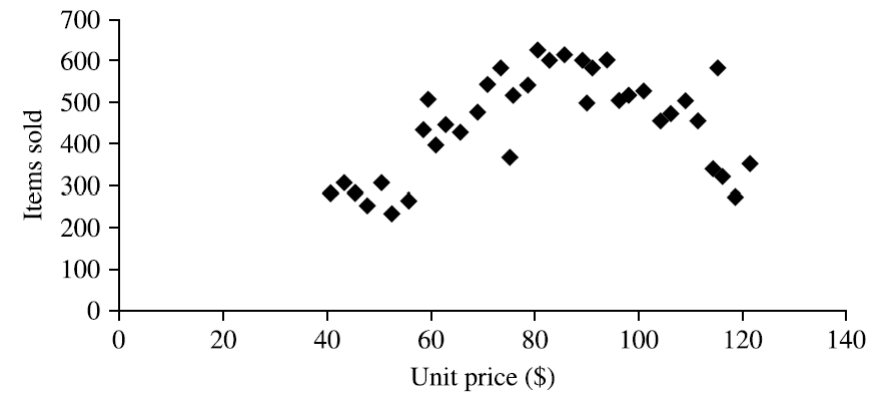


Fig: A scatter plot for tabulated data set.

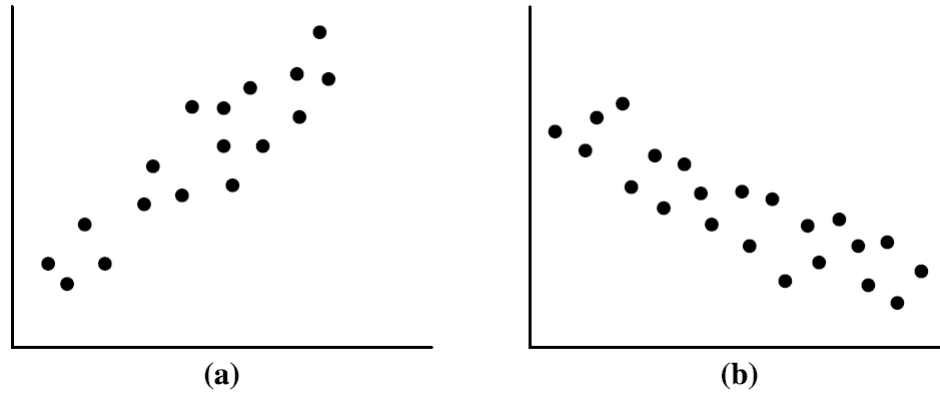


Fig: Scatter plots can be used to find (a) positive or (b) negative correlations between attributes.

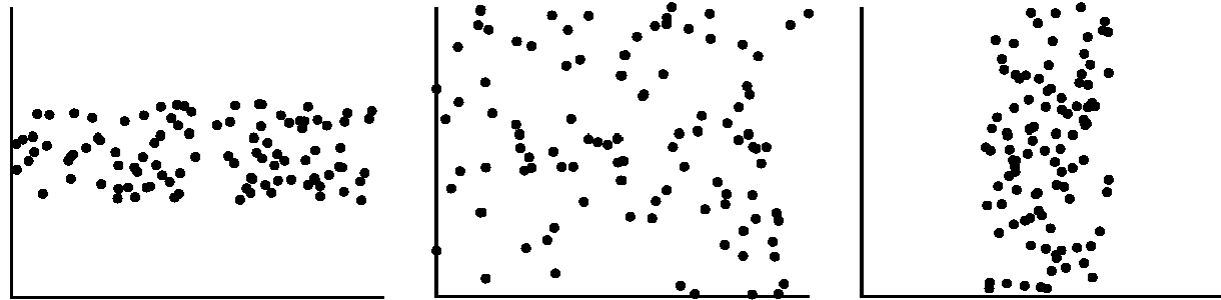


Fig: Three cases where there is no observed correlation between the two plotted attributes in each of the data sets.

Similarity and distance measures

- In data mining applications, such as clustering, outlier analysis, and nearest-neighbor classification, we need ways to assess how alike or unlike objects are in comparison to one another.
- To presents similarity and dissimilarity measures, which are referred to as measures of **proximity**.
- A similarity measure for two objects, i and j , will typically return value 0 if the objects are completely unlike. The higher the similarity value, the greater the similarity between objects. (Typically, a value of 1 indicates complete similarity, that is, the objects are identical.)
- A dissimilarity measure works the opposite way. It returns a value of 0 if the objects are the same (and therefore, far from being dissimilar). The higher the dissimilarity value, the more dissimilar the two objects are.

- **Real-Life Example Use-case** : In response to the shortage of medical resources during the rise of COVID-19 cases, engineers can use Machine Learning to assist in basic diagnosis.
- By applying clustering algorithms, patients can be grouped based on their symptoms into two clusters: COVID-19 positive and negative. New patients can then be diagnosed by comparing their symptoms with these clusters to predict infection.

Data matrix vs. dissimilarity matrix:

- The objects are $x_1 = (x_{11}, x_{12}, \dots, x_{1p})$, $x_2 = (x_{21}, x_{22}, \dots, x_{2p})$, and so on, where x_{ij} is the value for object x_i of the j th attribute. For brevity, we hereafter refer to object x_i as object i . The objects may be tuples in a relational database and are also referred to as **data samples or feature vectors**.
- Main memory-based clustering and nearest-neighbor algorithms typically operate on either of the following two data structures:

- **Data matrix** (or *object-by-attribute structure*): This structure stores the n data objects in the form of a relational table or an n -by- p matrix (n objects $\times p$ attributes): Each row corresponds to an object. As part of our notation, we may use f to index through the p attributes.

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}.$$

- **Dissimilarity matrix** (or *object-by-object structure*): This structure stores a collection of proximities that are available for all pairs of n objects. It is often represented by an n -by- n table: where $d(i, j)$ is the measured **dissimilarity** or “difference” between objects i and j .

$$\begin{bmatrix} 0 & & & & \\ d(2, 1) & 0 & & & \\ d(3, 1) & d(3, 2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n, 1) & d(n, 2) & \cdots & \cdots & 0 \end{bmatrix},$$

- Measures of similarity can often be expressed as a function of measures of dissimilarity. For example, for nominal data,

$$sim(i, j) = 1 - d(i, j),$$

Where, $sim(i, j)$ is the similarity between objects i and j .

Proximity Measures for Nominal Attributes:

- A nominal attribute can take on two or more states.
- For example, map color is a nominal attribute that may have, say, five states: red, yellow, green, pink, and blue.
- Let the number of states of a nominal attribute be M . The states can be denoted by letters, symbols, or a set of integers, such as $1, 2, \dots, M$. Notice that such integers are used just for data handling and do not represent any specific ordering.

“How is dissimilarity computed between objects described by nominal attributes?”

The dissimilarity between two objects i and j can be computed based on the ratio of mismatches:

$$d(i, j) = (p - m)/p \quad \text{---(1)}$$

where m is the number of *matches* (i.e., the number of attributes for which i and j are in the same state), and p is the total number of attributes describing the objects.

- **For example, Dissimilarity between nominal attributes.** Suppose that we have the sample data of Table 2.4, except that only the *object-identifier* and the attribute *test-1* are available, where *test-1* is nominal. (We will use *test-2* and *test-3* in later examples.)

Table 2.4 A sample data table containing attributes of mixed types.

Object Identifier	Test-1 (nominal)	Test-2 (ordinal)	Test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

Let's compute the dissimilarity matrix

$$\begin{bmatrix} 0 & & & \\ d(2, 1) & 0 & & \\ d(3, 1) & d(3, 2) & 0 & \\ d(4, 1) & d(4, 2) & d(4, 3) & 0 \end{bmatrix}.$$

- Since here we have one nominal attribute, *test-1*, we set $p = 1$ in Eq. (1) so that $d(i, j)$ evaluates to 0 if objects i and j match, and 1 if the objects differ. Thus, we get

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

From this, we see that all objects are dissimilar except objects 1 and 4 (i.e., $d(4, 1) = 0$).

Alternatively, similarity can be computed as

$$\text{sim}(i, j) = 1 - d(i, j) = m/p$$

Proximity measures for binary attributes:

- Recall that a binary attribute has only one of two states, 0 and 1, where 0 means that the attribute is absent, and 1 means that it is present.

“So, how can we compute the dissimilarity between two binary attributes?”

- One approach involves computing a dissimilarity matrix from the given binary data.
 - If all binary attributes are thought of as having the same weight, we have the 2×2 contingency table of table below,
 - where q is the number of attributes that equal 1 for both objects i and j ,
 - r is the number of attributes that equal 1 for object i but equal 0 for object j ,
 - s is the number of attributes that equal 0 for object i but equal 1 for object j ,
 - And t is the number of attributes that equal 0 for both objects i and j .
- The total number of attributes is p , where $p = q + r + s + t$.

Table 2.5 Contingency table for binary attributes.				
		Object j		
		1	0	sum
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
	sum	$q + s$	$r + t$	p

- Recall that for symmetric binary attributes, each state is equally valuable. Dissimilarity that is based on symmetric binary attributes is called **symmetric binary dissimilarity**. If objects i and j are described by symmetric binary attributes, then the dissimilarity between i and j is

$$d(i, j) = \frac{r + s}{q + r + s + t}.$$

- For asymmetric binary attributes, the two states are not equally important, such as the *positive* (1) and *negative* (0) outcomes of a disease test. Given two asymmetric binary attributes, the agreement of two 1s (a positive match) is then considered more significant than that of two 0s (a negative match).
- Therefore, such binary attributes are often considered “monary” (having one state). The dissimilarity based on these attributes is called **asymmetric binary dissimilarity**, where the number of negative matches, t , is considered unimportant and is thus ignored in the following computation:

$$d(i, j) = \frac{r + s}{q + r + s}.$$

- For example, the **asymmetric binary similarity** between the objects i and j can be computed as

$$sim(i, j) = \frac{q}{q + r + s} = 1 - d(i, j).$$

The coefficient $sim(i, j)$ is called the **Jaccard coefficient**

- **For example**, Dissimilarity between binary attributes. Suppose that a patient record table (Table 2.4) contains the attributes name, gender, fever, cough, test-1, test-2, test-3, and test-4, where name is an object identifier, gender is a symmetric attribute, and the remaining attributes are asymmetric binary.

Table 2.6 Relational table where patients are described by binary attributes.							
Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

- For asymmetric attribute values, let the values Y (yes) and P (positive) be set to 1, and the value N (no or negative) be set to 0. Suppose that the distance between objects (patients) is computed based only on the asymmetric attributes.
- According to **asymmetric binary dissimilarity**, the distance between each pair of the three patients—Jack, Mary, and Jim—is

$$d(Jack, Jim) = \frac{1 + 1}{1 + 1 + 1} = 0.67,$$

$$d(Jack, Mary) = \frac{0 + 1}{2 + 0 + 1} = 0.33,$$

$$d(Jim, Mary) = \frac{1 + 2}{1 + 1 + 2} = 0.75.$$

- These measurements suggest that Jim and Mary are unlikely to have a similar disease because they have the highest dissimilarity value among the three pairs. Of the three patients, Jack and Mary are the most likely to have a similar disease.

Dissimilarity of Numeric Data: Minkowski Distance

- Distance measures that are commonly used for computing the dissimilarity of objects described by numeric attributes. These measures include the Euclidean, Manhattan, and Minkowski distances.

Euclidean distances:

- The most popular distance measure is **Euclidean distance** (i.e., straight line or “as the crow flies”). Let $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ be two objects described by p numeric attributes.
- The Euclidean distance between objects i and j is defined as

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}.$$

Manhattan (or city block) distance:

- Another well-known measure is the **Manhattan (or city block) distance**, named so because it is the distance in blocks between any two points in a city (such as 2 blocks down and 3 blocks over for a total of 5 blocks). It is defined as

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|.$$

- Both the Euclidean and the Manhattan distance satisfy the following mathematical properties:
 - **Nonnegativity:** $d(i, j) \geq 0$: Distance is a nonnegative number.
 - **Identity of indiscernibles:** $d(i, i) = 0$: The distance of an object to itself is 0.
 - **Symmetry:** $d(i, j) = d(j, i)$: Distance is a symmetric function.
 - **Triangle inequality:** $d(i, j) \leq d(i, k) + d(k, j)$: Going directly from object i to object j in space is no more than making a detour over any other object k . A measure that satisfies these conditions is known as **metric**.

Minkowski distance:

- **Minkowski distance** is a generalization of the Euclidean and Manhattan distances. It is defined as

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h},$$

Where, h is a real number such that $h \geq 1$. It represents the Manhattan distance when $h = 1$ (i.e., $L1$ norm) and Euclidean distance when $h = 2$ (i.e., $L2$ norm).

Supremum distance:

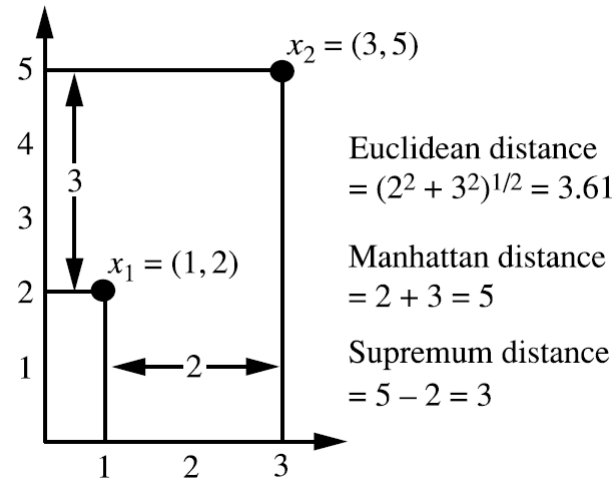
- The **supremum distance** (also referred to as ***L*max**, ***L*[∞] norm**, and the **Chebyshev distance**) is a generalization of the Minkowski distance for $h \rightarrow \infty$.
- To compute it, we find the attribute f that gives the maximum difference in values between the two objects. This difference is the supremum distance, defined more formally as:

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|.$$

- The L^∞ norm is also known as the *uniform norm*.

For example, Euclidean distance and Manhattan distance. Let $x_1 = (1, 2)$ and $x_2 = (3, 5)$ represent two objects as shown in Fig below.

- The Euclidean distance between the two is $\sqrt{2^2 + 3^2} = 3.61$.
- The Manhattan distance between the two is $2 + 3 = 5$.



- Let's use the same two objects, $x_1 = (1, 2)$ and $x_2 = (3, 5)$, as in Fig above. The second attribute gives the greatest difference between the values for the objects. That is, $\max\{(|3 - 1|, |5 - 2|)\} = 3$. This is the supremum distance between the two objects.
- This is the supremum distance between both objects. If each attribute is assigned a weight according to its perceived importance, the **weighted Euclidean distance can be computed as**

$$d(i, j) = \sqrt{w_1|x_{i1} - x_{j1}|^2 + w_2|x_{i2} - x_{j2}|^2 + \cdots + w_m|x_{ip} - x_{jp}|^2}.$$

Data quality, data cleaning, and data integration

- Data Preprocessing: An Overview,
 - **Data Quality,**
 - **Major Tasks in Data Preprocessing**

Why Preprocess the Data?

- **Preprocessing** of data is mainly to check the data quality.
- Measures for data quality: A multidimensional views,
 - **Accuracy:** To check whether the data entered is correct or not.
 - **Completeness:** To check whether the data is available or not recorded.
 - **Consistency:** To check whether the same data is kept in all the places that do or do not match.
 - **Timeliness:** The data should be updated correctly.
 - **Believability:** The data should be trustable.
 - **Interpretability:** The understandability of the data.

Major Tasks in Data Preprocessing

- Data goes through a series of steps during pre processing:
- **Data Cleaning:** Data is cleaned through processes such as filling in missing values, smoothing the noisy data, or resolving the inconsistencies in the data.
- **Data Integration:** Data with different representations are put together and conflicts within the data are resolved.
- **Data Transformation:** Data is normalized, aggregated and generalized.
- **Data Reduction:** This step aims to present a reduced representation of the data in a data warehouse.
- **Data Discretization:** Involves the reduction of a number of values of a continuous attribute by dividing the range of attribute intervals.

Data cleaning

- Real-world data tend to be incomplete, noisy, and inconsistent.
- **Data cleaning** (or *data cleansing*) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.
- **Missing Data:** This situation arises when some data is missing in the data. It can be handled in various ways. Some of them are:
 - **Ignore the tuples:** This is usually done when the class label is missing (assuming the mining task involves classification). This method is not very effective, unless the tuple contains several attributes with missing values.
 - **Fill in the missing value manually:** In general, this approach is time consuming and may not be feasible given a large data set with many missing values.

- **Use a global constant to fill in the missing value:** Replace all missing attribute values by the same constant such as a label like “*Unknown*” or $-\infty$.
- **Use the attribute mean or median for all samples belonging to the same class as the given tuple:** For example, if classifying customers according to *credit risk*, we may replace the missing value with the mean *income* value for customers in the same credit risk category as that of the given tuple.
- **Use the most probable value to fill in the missing value:** This may be determined with regression, inference-based tools using a Bayesian formalism, or decision tree induction.
- For example, using the other customer attributes in your data set, you may construct a decision tree to predict the missing values for *income*.

- **Noisy Data:** “*What is noise?*” **Noise** is a random error or variance in a measured variable. or meaningless data. e.g. Salary=”-10” .
- Incorrect attribute values may be due to:
 - Faulty data collection instruments.
 - Data entry problem.
 - Data transmission problem .
 - Technology limitation.
 - Inconsistency in naming convention.
 - Duplicate records, incomplete data(e.g. Occupation = “ ” (missing data)), inconsistent data (e.g. Was rating “1, 2, 3”, now “A, B, C”)

How to Handle Noisy Data ?

- i. **Binning Method:** Binning methods smooth a sorted data value by consulting its “neighborhood,” that is, the values around it. The sorted values are distributed into a number of “buckets,” or *bins*.
 - Because binning methods consult the neighborhood of values, they perform *local* smoothing, then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
 - In **smoothing by bin means**, each value in a bin is replaced by the mean value of the bin. For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore, each original value in this bin is replaced by the value 9.
 - **Smoothing by bin medians** can be employed, in which each bin value is replaced by the bin median.
 - **Smoothing by bin boundaries**, the minimum and maximum values in a given bin are identified as the *bin boundaries*. Each bin value is then replaced by the closest boundary value.
 - Binning is also used as a discretization technique.

Example:

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Explanation:

- The data for *price* are first sorted and then partitioned into *equal-frequency* bins of size 3.

Smoothing by bin means:

Bin 1($27/3=9$): $4+8+15=27$

Bin 2 ($66/3=22$) : $21+21+24=66$

Bin 3 ($87/3=29$) : $25+28+34=87$

Smoothing by bin boundaries:

Bin1: 4, 8,15 boundary is (4,15) where 8 is closure with 4 than 15

Bin 2: 21,21,24 boundary is (21,24) where 21 is closure with 21 and 24 are closure with 24

Bin3:25,25,34 boundary is (25,34) where 28 is closure with 25 than 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Smoothing by bin means:

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

Smoothing by bin boundaries:

Bin 1: 4, 4, 15

Bin 2: 21, 21, 24

Bin 3: 25, 25, 34

Figure : Binning methods for data smoothing.

ii. Regression: Data smoothing can also be done by regression, a technique that conforms data values to a function.

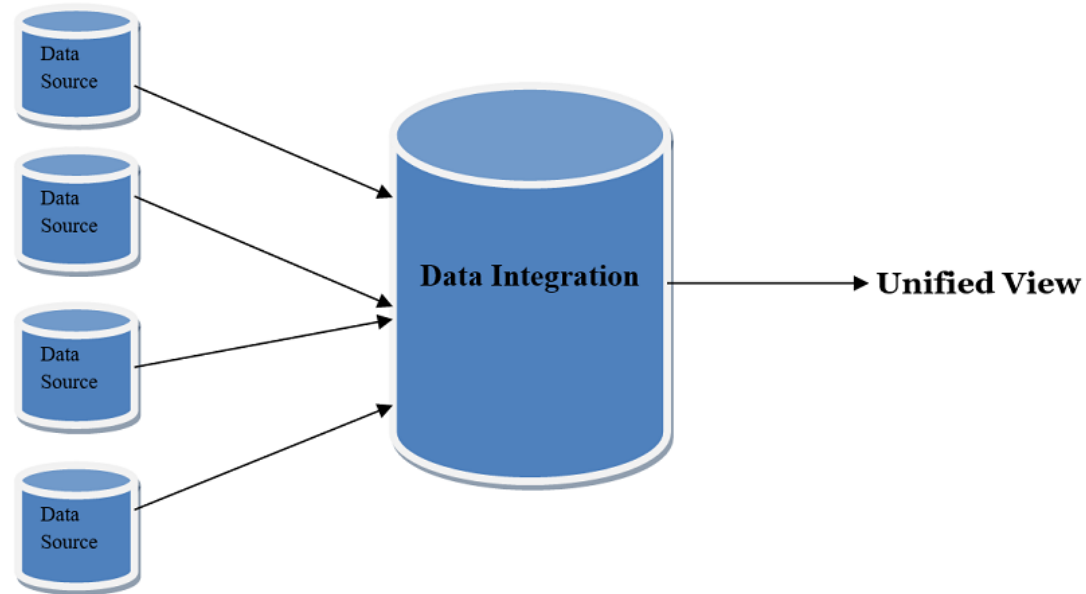
- **Linear regression** involves finding the “best” line to fit two attributes (or variables) so that one attribute can be used to predict the other.
- **Multiple linear regression** is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

iii. Outlier analysis: Outliers may be detected by clustering, for example, where similar values are organized into groups, or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered outliers.

- Data inconsistencies occur when information about the same entity or attribute is different across various sources, systems, or formats

Data Integration

- Integration of multiple databases or data files into the single store of identical data is known as Data Integration.
- Data Integration is usually performed to create data sets for machine learning algorithms and to predict the statistical information from the data during the data mining.



- **Data mining often requires data integration**—the merging of data from multiple data stores. Careful integration can help reduce and avoid redundancies and inconsistencies in the resulting data set.
- This can help improve the accuracy and speed of the subsequent data mining process.
- We integrate data from various resources like banking transactions, invoices, customer records, Twitter, blog postings, image, audio or video data, electronic data interchange (EDI) files, spreadsheets, and sensor data.
- E.g. Integrated data from various patient records and clinics assist clinicians in identifying medical disorders and diseases by integrating data from many systems into a single perspective of beneficial information from which useful insights can be derived.

Tight Coupling:

- Tight Coupling is a process of combining data from various data sources using ETL (Extraction, Transformation, and Loading) into a single storage system such as Data Warehouse.
 - **Centralized Repository:** Data is brought into a unified data warehouse.
 - **Integration Mechanism:** Integration happens at larger levels like involving entire datasets or schemas.
 - **Process Involved:** The ETL processes – Extraction, Transformation, and Loading – are used for data collection and integration.

Benefits of Tight Coupling:

- Provides **faster query execution** since data is already aggregated.
 - Ensures **consistency and standardization** by storing data in a common format.
 - Enables **comprehensive analytics and reporting** from a unified data source.
-
- This approach is also known as data warehousing, and it enables data consistency and integrity, but it can be inflexible and difficult to change or update.

Loose Coupling:

- Loose coupling relies on middleware, data virtualization, or federated databases to provide on-demand access to distributed data sources without physically moving or storing the data in one location.
- Instead of creating a centralized repository, this approach enables data access from different systems dynamically.

Benefits of Loose Coupling:

- **Greater flexibility** since data remains in its original source.
- **Scalability** by allowing integration across multiple systems without extensive restructuring.
- Reduces **storage costs** as data is not duplicated into a central repository.

Issues in Data Integration

Entity Identification Problem:

- identify real world entities from multiple data sources.
- For example, how can a data analyst or a computer be sure that *customer_id* in one database and *cust_number* in another refer to the same attribute? Moreover, metadata may be used to help entity identification (e.g., data codes for *pay_type* in one database may be “H” and “S” but 1 and 2 in another).
- Examples of metadata for each attribute include the name, meaning, data type, range of values permitted for the attribute, and null rules for handling blank, zero, or null values. Such metadata can be used to help avoid errors in schema integration.

- **Redundancy and Correlation Analysis:** An attribute (such as *annual revenue*, for instance) may be redundant if it can be “derived” from another attribute or set of attributes. Inconsistencies in attribute or dimension naming can also cause redundancies in the resulting data set. *Redundancy* is another important issue in data integration.
- An attribute (such as *annual revenue*, for instance) may be redundant if it can be “derived” from another attribute or set of attributes.
- Inconsistencies in attribute or dimension naming can also cause redundancies in the resulting data set.
- Some redundancies can be detected by **correlation analysis**. Given two attributes, such analysis can measure how strongly one attribute implies the other, based on the available data.
- For nominal data, we can use the χ^2 (*chi-square*) test. For numeric attributes, we can use the *correlation coefficient* and *covariance*, both of which assess how one attribute’s values vary from those of another.

- **Tuple Duplication:** In addition to detecting redundancies between attributes, duplication should also be detected at the tuple level (e.g., where there are two or more identical tuples for a given unique data entry).
- For example, if a purchase order database contains attributes for the purchaser's name and address instead of a key to this information in a purchaser database, discrepancies can occur, such as the same purchaser's name appearing with different addresses within the purchase order database.
- **Data Value Conflict Detection and Resolution:** Data integration also involves the *detection and resolution of data value conflicts*. For example, for the same real-world entity, attribute values from different sources may differ. This may be due to differences in representation, scaling, or encoding.
- For instance, a *weight* attribute may be stored in metric units in one system and British imperial units in another.
- For a hotel chain, the *price* of rooms in different cities may involve not only different currencies but also different services (e.g., free breakfast) and taxes.

Data Transformation

- The data are transformed or consolidated so that the resulting mining process may be more efficient, and the patterns found may be easier to understand.

Data Transformation Strategies Overview:

1. **Smoothing**, which works to remove noise from the data. Techniques include binning, regression, and clustering.
 2. **Attribute construction** (or *feature construction*), where new attributes are constructed and added from the given set of attributes to help the mining process.
 3. **Aggregation**, where summary or aggregation operations are applied to the data.
- For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts. This step is typically used in constructing a data cube for data analysis at multiple abstraction levels.

4. Normalization, where the attribute data are scaled so as to fall within a smaller range, such as -1.0 to 1.0, or 0.0 to 1.0.

- There are many methods for data normalization. We study *min-max normalization*, *z-score normalization*, and *normalization by decimal scaling*. For our discussion, let A be a numeric attribute with n observed values, $v_1, v_2, \dots, v_n$.
- **Min-max normalization** performs a linear transformation on the original data. Suppose that $\min_A$ and $\max_A$ are the minimum and maximum values of an attribute, A .
- Min-max normalization maps a value, v_i , of A to v'_i in the range $[\text{new_min}_A, \text{new_max}_A]$ by computing.

$$v'_i = \frac{v_i - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

Example: Suppose that the minimum and maximum values for the attribute income are \$12,000 and \$98,000, respectively. We would like to map income to the range [0.0, 1.0]. By min-max normalization, a value of \$73,600 for income is transformed to

$$\frac{73,600 - 12,000}{98,000 - 12,000}(1.0 - 0) + 0 = 0.716$$

- In **Z-score normalization** (or zero-mean normalization), the values for an attribute, A, are normalized based on the mean (i.e., average) and standard deviation of A. A value, v_i , of A is normalized to v_i' by computing

$$v_i' = (v_i - A') / \delta_A$$
 where A' and δ_A are the mean and standard deviation, respectively, of attribute A.

Example: Suppose that the mean and standard deviation of the values for the attribute *income* are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for *income* is transformed to $(73,600 - 54,000) / 16,000 = 1.225$

- **Normalization by decimal scaling** normalizes by moving the decimal point of values of attribute A . The number of decimal points moved depends on the maximum absolute value of A . A value, v_i , of A is normalized to v_i' by computing

$$v_i' = v_i / 10^j \quad \text{where } j \text{ is the smallest integer such that } \max(|v_i'|) < 1.$$

Example: Suppose that the recorded values of A range from -986 to 917. The maximum absolute value of A is 986. To normalize by decimal scaling, we therefore divide each value by 1000 (i.e., $j=3$) so that -986 normalizes to -0.986 and 917 normalizes to 0.917.

5. Discretization, where the raw values of a numeric attribute (e.g., *age*) are replaced by interval labels (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*).

- The labels, in turn, can be recursively organized into higher-level concepts, resulting in a *concept hierarchy* for the numeric attribute.

Data Reduction

- **Data reduction** techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data. i.e. , mining on the reduced data set should be more efficient yet produce the same (or almost the same) analytical results.

Overview of Data Reduction Strategies

- Data reduction strategies includes:
 1. **Dimensionality reduction**, is the process of reducing the number input variables or attributes under consideration.
- Methods include
- *wavelet transforms* and *principal components analysis*, which transform or project the original data onto a smaller space.

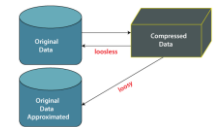
- ***Attribute subset selection*** is a method of dimensionality reduction in which irrelevant, weakly relevant, or redundant attributes or dimensions are detected and removed .

2. Numerosity reduction, techniques replace the original data volume by alternative, smaller forms of data representation. These techniques may be parametric or nonparametric.

- For **parametric methods**, a model is used to estimate the data, so that typically only the data parameters need to be stored, instead of the actual data.-(Outliers may also be stored.)
- Regression and log-linear models are examples.
- **Nonparametric methods** for storing reduced representations of the data include *histograms*, *clustering* , *sampling* , and *data cube aggregation*.

3. Data compression, transformations are applied so as to obtain a reduced or “compressed” representation of the original data.

- If the original data can be reconstructed from the compressed data without any information loss, the data reduction is called **lossless**.
- *Algorithms*: Run Length Encoding, Lempel-Ziv-Welch, Huffman Coding, Arithmetic encoding etc.)
- If, instead, we can reconstruct only an approximation of the original data, then the data reduction is called **lossy**. (Algorithms : Transform coding, Discrete Cosine Transform, Discrete Wavelet Transform, fractal compression etc.)



- There are several lossless algorithms for string compression; however, they typically allow only limited data manipulation.
- Dimensionality reduction and numerosity reduction techniques can also be considered forms of data compression.

Data discretization, Concept Hierarchy Generation

- Data discretization is a common data transformation technique, where the raw values of a numeric attribute (e.g., *age*) are replaced by interval labels (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*).
- The labels, in turn, can be recursively organized into higher-level concepts, resulting in a ***concept hierarchy*** for the numeric attribute.

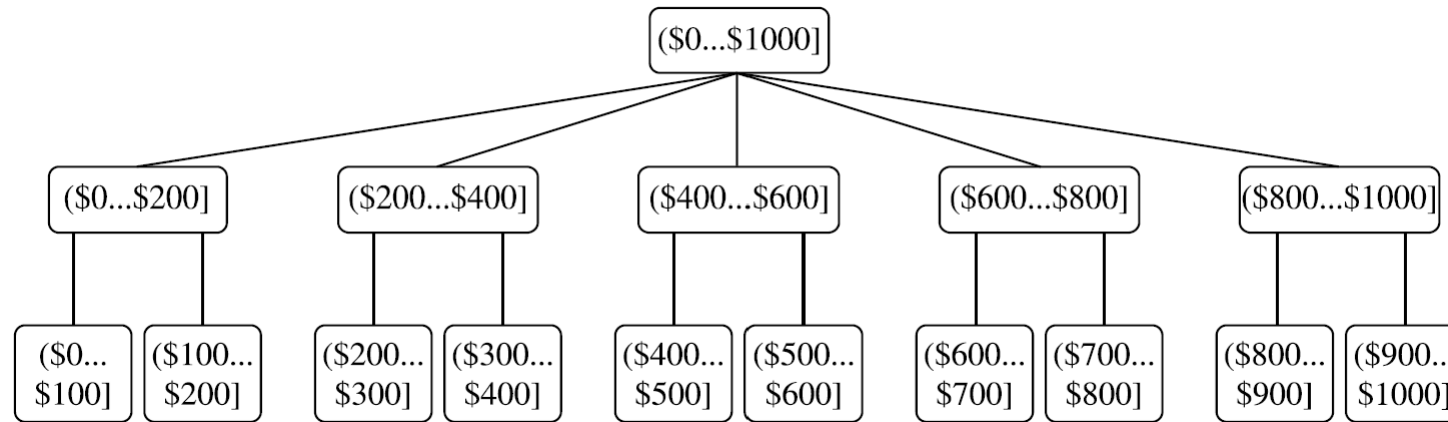


Fig. : Shows a concept hierarchy for the attribute *price*.

- Discretization techniques can be categorized based on how the discretization is performed, such as whether it uses **class information** or which direction it proceeds (i.e., **top-down vs. bottom-up**).
- If the discretization process uses class information, then we say it is ***supervised discretization***. Otherwise, it is ***unsupervised***.
- If the process starts by first finding one or a few points (called *split points* or *cut points*) to split the entire attribute range and then repeats this recursively on the resulting intervals, it is called ***top-down discretization or splitting***.
- This contrasts with *bottom-up discretization* or *merging*, which starts by considering all of the continuous values as potential split-points, removes some by merging neighborhood values to form intervals, and then recursively applies this process to the resulting intervals.

Concept Hierarchy Generation

- ***Concept hierarchy*** organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse
- Concept hierarchies facilitate *drilling and rolling* in data warehouses to view data in multiple granularity.
- ***Concept hierarchy formation:*** Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for *age*) by higher level concepts (such as *child*, *young*, mature or *old*).
- Concept hierarchies can be explicitly specified by domain experts and/or data warehouse designers.
- Concept hierarchy can be automatically formed for both numeric and nominal data.

- Example : Suppose we have an attribute of Age with the given values

Before Discretization

Age : 1, 5, 9, 4, 7, 11, 14, 17, 13, 18, 19, 31, 33, 36, 42, 44, 46, 70, 74, 78, 77

Attribute	Age	Age	Age	Age
	1,5,4,9,7	11,14,17,13,18,19	31,33,36,42,44,46	70,74,77,78
After Discretization	Child	Young	Mature	Old

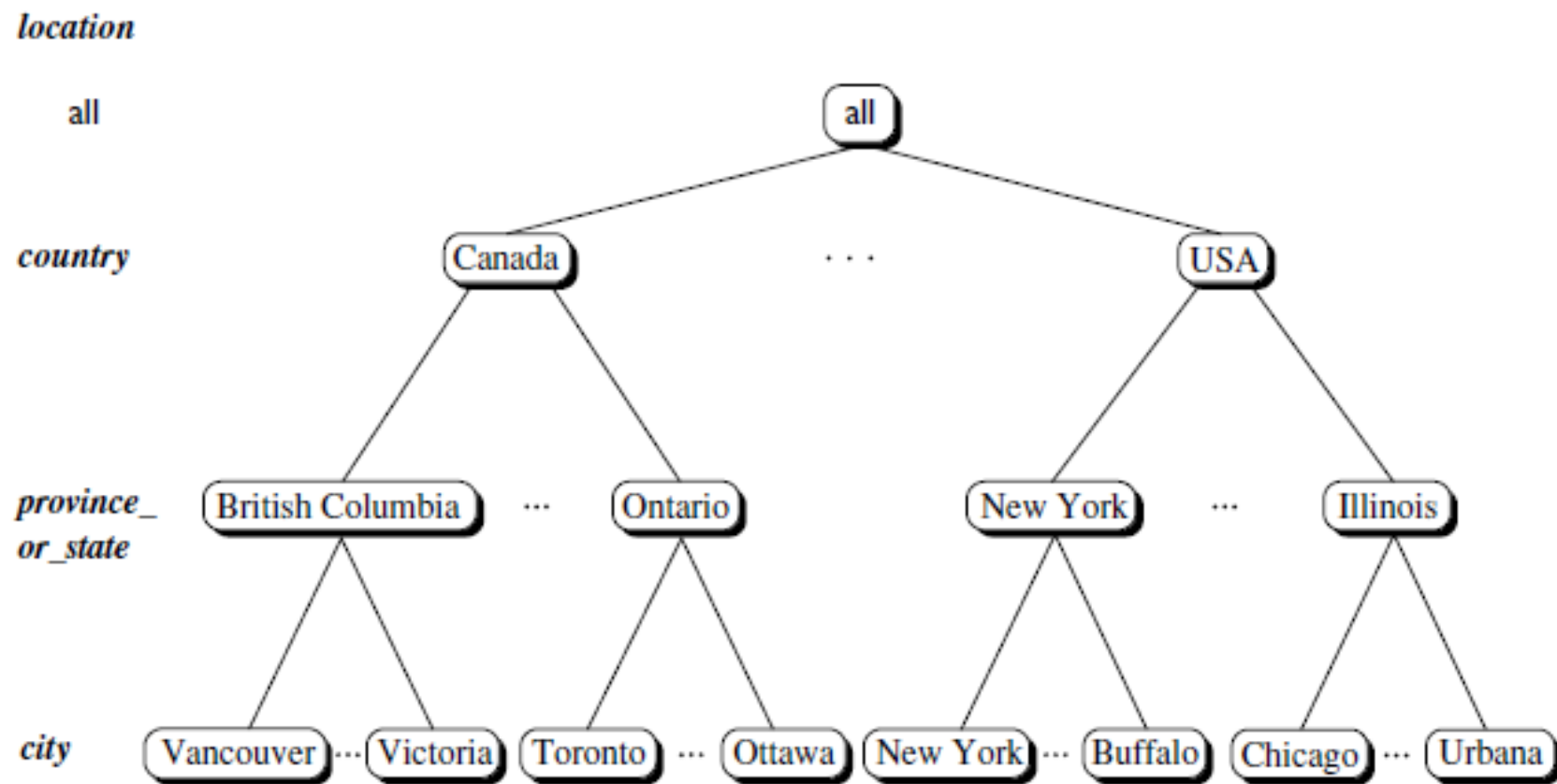


Figure: A concept hierarchy for *location*.

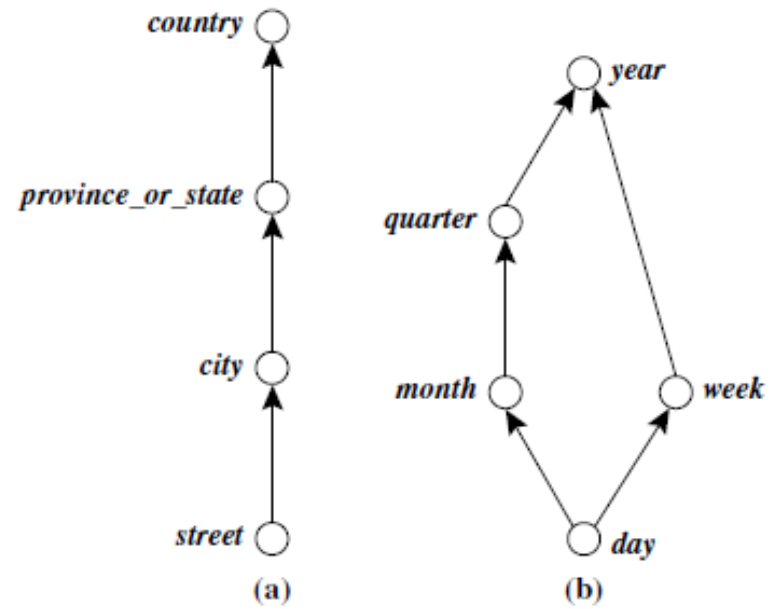


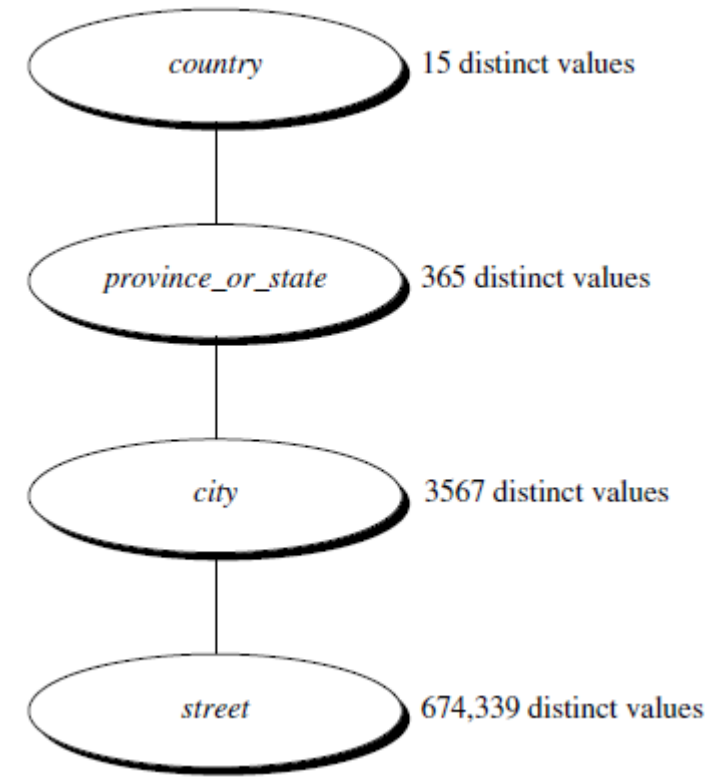
Figure: Hierarchical and lattice structures of attributes in warehouse dimensions: (a) a hierarchy for *location* and (b) a lattice for *time*.

Concept Hierarchy Generation for Nominal Data

- Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts
 - E.g., $street < city < state < country$
- Specification of a hierarchy for a set of values by explicit data grouping
 - E.g., $\{KTM, Lalitpur, Bhaktapur\} < Bagmati$
- Specification of only a partial set of attributes
 - E.g., only $street < city$, not others
- Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values
 - E.g., for a set of attributes: $\{street, city, state, country\}$

Automatic Concept Hierarchy Generation

- Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
- The attribute with the most distinct values is placed at the lowest level of the hierarchy
- Exceptions, e.g., weekday, month, quarter, year



- Two basic discretization techniques,
 - **Binning and**
 - **Histogram analysis.**
- **Discretization by binning:**
- Binning is a top-down splitting technique based on a specified number of bins.
- For example, attribute values can be discretized by applying equal-width or equal-frequency binning and then replacing each bin value by the bin mean or median, as in *smoothing by bin means* or *smoothing by bin medians*, respectively.
- **Discretization by histogram analysis:**
- Histogram analysis is an unsupervised discretization technique because it does not use class information.

Example: The numbers have been sorted: 1, 1, 5, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.

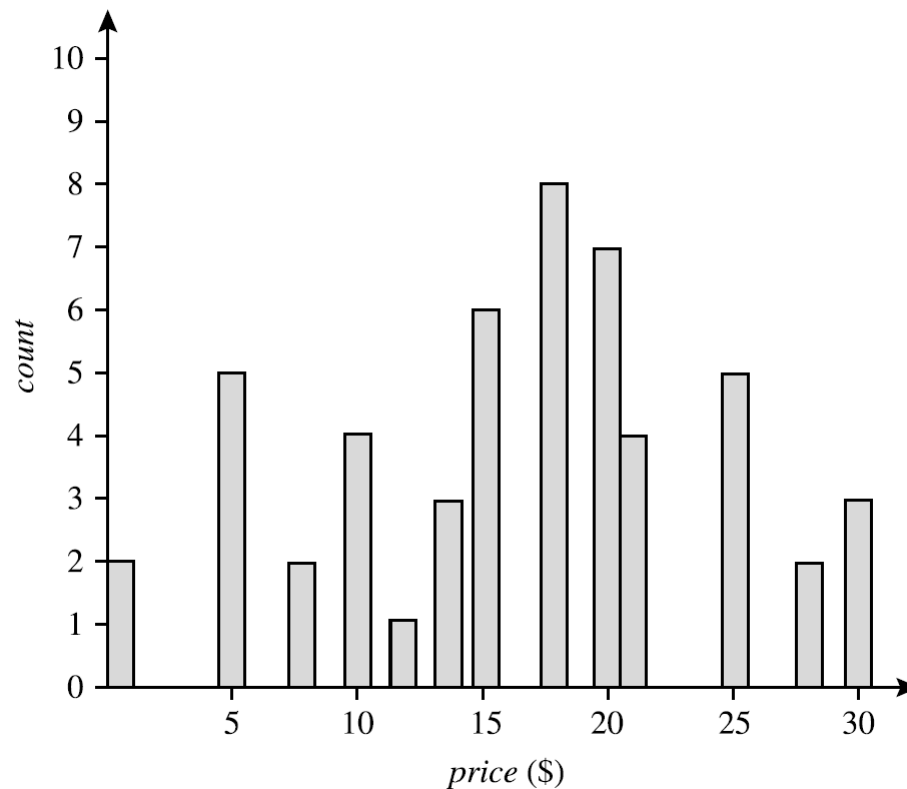


Fig: A histogram for *price* using singleton buckets—each bucket represents one price–value/frequency pair.

Advanced Discretization Techniques:

- **Cluster Analysis:** Data points are grouped into clusters using clustering algorithms (e.g., k-means), and each cluster represents a discrete category.
- **Decision Tree Analysis:** Decision trees (e.g., ID3, C4.5) can be used to find cut points that best split the data based on class labels, leading to supervised discretization.
- **Correlation Analysis:** This method examines relationships between attributes and identifies discretization points that preserve correlations or dependencies among variables.

Dimensionality Reduction

- **Dimensionality reduction** is the process of reducing the number of random variables or attributes or features under consideration.
 - It is a crucial step in data preprocessing, especially when dealing with high-dimensional data, as it helps in improving model performance, reducing overfitting, and enhancing visualization.
- Dimensionality reduction methods,
- Linear Methods:
 - **Principal components analysis (PCA)**- It transforms or projects the original data onto a smaller space.
 - **Attribute subset selection**- in which irrelevant, weakly relevant, or redundant attributes or dimensions are detected and removed.
 - Nonlinear methods:
 - **kernel PCA**
 - **Stochastic neighbor embedding**

Principal components analysis (PCA):

- Also called the Karhunen-Loeve, or K-L, method) searches for k d -dimensional orthonormal vectors that can best be used to represent the data, where $k \leq d$. The original data are thus projected onto a much smaller space, resulting in dimensionality reduction.
- The basic procedure is as follows:

1. Normalization:

- Each attribute is normalized to fall within the same range.
- Prevents attributes with larger numeric ranges from dominating those with smaller ranges.

2. Principal Component Computation:

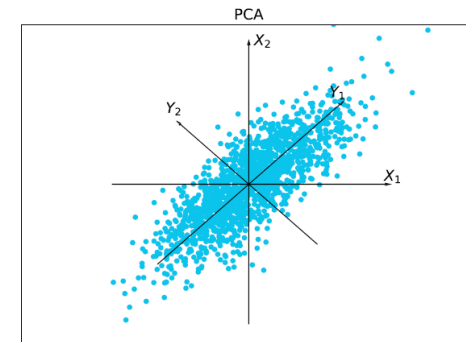
- Compute the covariance matrix of the normalized data.
- Find the eigenvectors and eigenvalues of the covariance matrix.
- The eigenvectors are orthonormal (unit length and mutually perpendicular).
- These form the principal components — a new basis for the data.
- The original data can now be expressed as a linear combination of these components.

3. Sorting by Significance (Variance):

- Principal components are sorted in descending order of their corresponding eigenvalues.
- Each component indicates the amount of variance it explains:
 - The first component captures the maximum variance in the data.
 - The second component captures the next highest variance, and so on.
- For example, in a 2D space (X_1 , X_2), the first two principal components (Y_1 , Y_2) can reveal structure in the data (as illustrated in diagrams like Fig. 2.17).

4. Dimensionality Reduction:

- Discard weaker components (those with small variance).
- Retain only the top k components to reduce data size.
- This preserves the most significant features of the data.
- A good approximation of the original data can often be reconstructed from these strong components.



Attribute subset selection:

- Reduces the data set size by removing irrelevant or redundant attributes (or dimensions).
- This makes mining focused on the relevant dimensions. Mining on a reduced set of attributes has an additional benefit:
- It reduces the number of attributes appearing in the discovered patterns, helping to make the patterns easier to understand.
- Heuristic methods that explore a reduced search space are commonly used for attribute subset selection.
- These methods are typically **greedy** in that, while searching through attribute space, they always make what looks to be the best choice at the time.
- locally optimal choice -globally good solution.

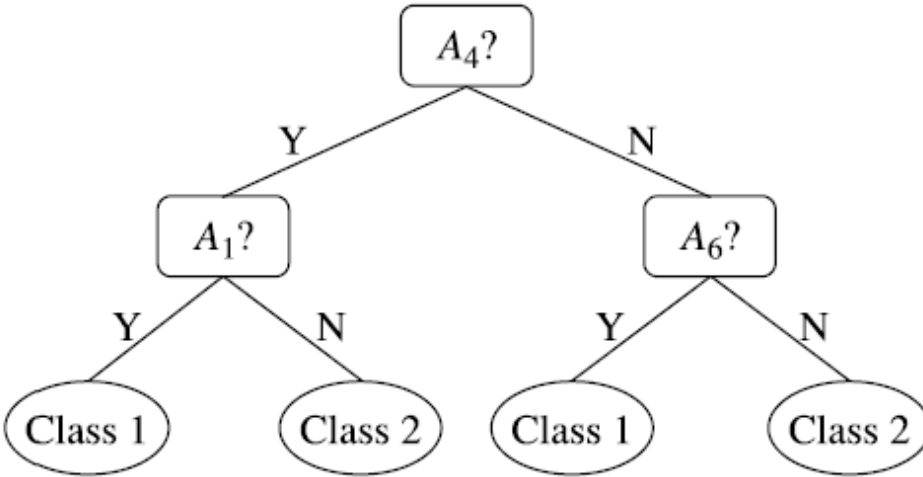
Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  <pre> graph TD A4["A4?"] -- Y --> A1["A1?"] A4 -- N --> A6["A6?"] A1 -- Y --> C1_1(["Class 1"]) A1 -- N --> C2_1(["Class 2"]) A6 -- Y --> C1_2(["Class 1"]) A6 -- N --> C2_2(["Class 2"]) </pre> $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

Fig: Greedy (heuristic) methods for attribute subset selection

- Basic heuristic methods of attribute subset selection include the following techniques,

1. Stepwise forward selection:

- The procedure starts with an empty set of attributes as the reduced set. The best of the original attributes is determined and added to the reduced set.
- At each subsequent iteration or step, the best of the remaining original attributes is added to the set.

2. Stepwise backward elimination:

- The procedure starts with the full set of attributes.
- At each step, it removes the worst attribute remaining in the set.

3. Combination of forward selection and backward elimination:

- The stepwise forward selection and backward elimination methods can be combined so that, at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.

4. Decision tree induction:

- Decision tree algorithms (e.g., ID3, C4.5, and CART) were originally intended for classification.
- Decision tree induction constructs a flowchart-like structure where each internal (nonleaf) node denotes a test on an attribute, each branch corresponds to an outcome of the test, and each external (leaf) node denotes a class prediction.

Nonlinear dimensionality reduction methods:

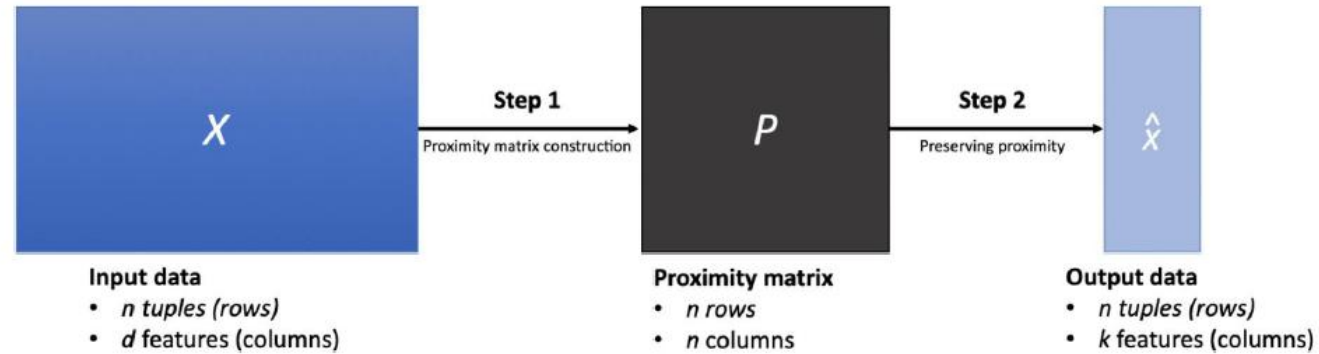


Table 2.8 Comparison of KPCA and SNE.

	Step 1: Proximity Construction	Step 2: Preserving Proximity
KPCA	$P(i, j) = \kappa(x_i, x_j)$	$\min \sum_{i,j=1}^n (P(i, j) - \hat{P}(i, j))^2 = \ P - \hat{P}\ _{fro}^2$
SNE	$P(i, j) = \frac{e^{-d_{ij}^2}}{\sum_{l=1, l \neq i}^n e^{-d_{il}^2}}$	$\min \sum_{i=1}^n \text{KL}(P_i \hat{P}_i)$

Fig: An illustration of nonlinear dimensionality reduction.

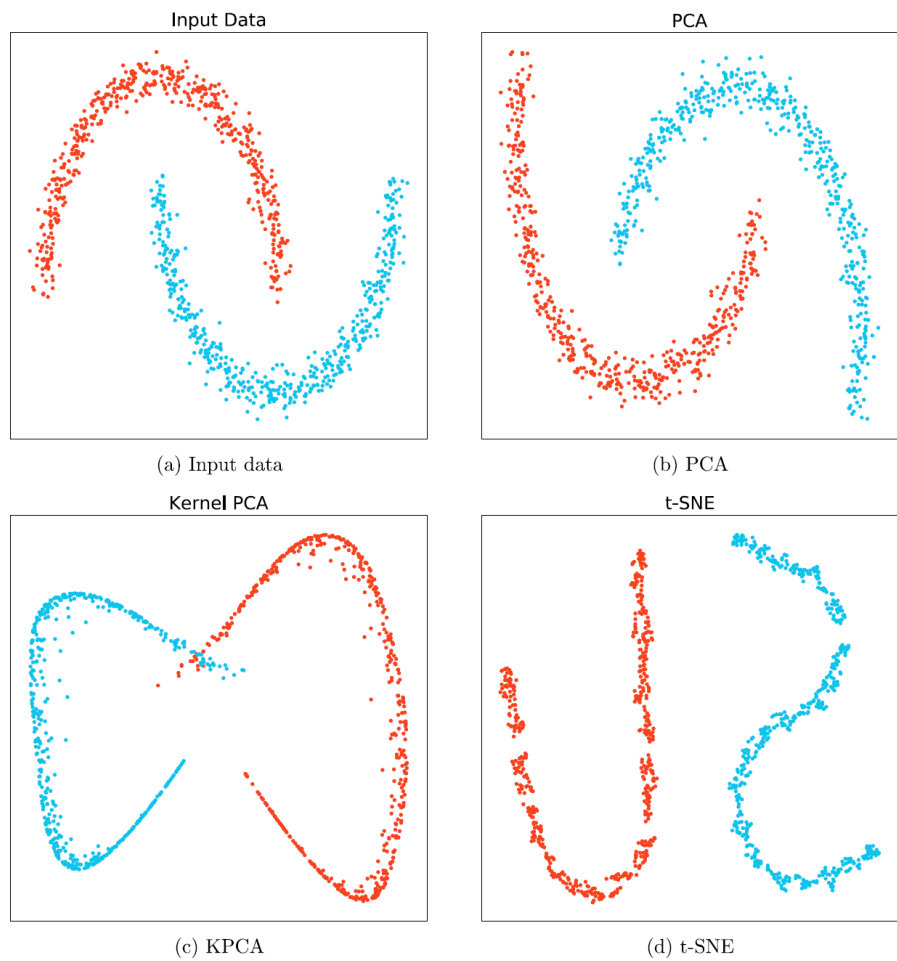


Fig: An example of linear vs. nonlinear dimensionality reduction methods.

- Another type of data reduction technique is called **numerosity reduction**, which uses parametric or nonparametric models to obtain smaller representations of the original data.
- Parametric models store only the model parameters instead of the actual data. Examples include regression and log-linear models.
- Nonparametric methods include histograms, clustering, sampling, and data cube aggregation.